

History-Conditioned Control in Open Systems

Teleonomic Potentials, Dissipative Computation, and Operational Criteria for Conditional Agency

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Abstract

Purpose-like descriptions are common in biology, neuroscience, robotics, and artificial intelligence, but the same language is often applied to systems with radically different capacities. A falling object follows a gradient, a thermostat corrects a measured error, a cell changes its behavior after stress, and a planning agent compares alternatives. Dissipation and stabilization alone do not make these systems equivalent. This paper develops a bounded mathematical framework for distinguishing four evidential levels: physical dissipation, regulation, history-conditioned adaptive control, and conditional agency. The core proposal is that a system becomes usefully teleonomic when internally accessible history changes its response to otherwise matched present conditions, returned consequences update that history, and the resulting policy differences survive matched null models and causal interventions. Agency requires additional evidence concerning individuality, system-relative normativity, interactional asymmetry, and a counterfactual action repertoire.

The framework replaces a unit-mixing scalar agency index with a claim vector. It separates observational routing measures, such as conditional mutual information and transfer entropy, from interventional causal effects. It also distinguishes a computational mismatch certificate from physical entropy production. For a declared mismatch functional, the balance law is written as mismatch reduction plus uncontrolled production; for a thermodynamically resolved stochastic process, entropy production is defined by a forward-to-reverse path-probability ratio. Neither equation, by itself, establishes purpose or agency. The paper specifies state-only, shuffled-history, consequence-permuted, capacity-matched, energy-matched, and post-hoc narrative nulls; proposes synthetic-controller, morphogenetic, and neural test programs; and states explicit failure conditions. A deterministic five-system reference benchmark then tests the evidential ladder. Open-loop and memoryless-feedback systems fail the history package; writable-history and planner systems pass their declared diagnostic levels; and an observer-only record improves prediction while failing the consequence-writing and memory-intervention tests. The benchmark validates the decision logic, not biological agency. The paper preserves the teleonomic / Telos research bridge developed in *Super Information Theory* while removing universal-purpose, cross-domain coefficient, and ordinary-buoyancy overclaims. The result is a substrate-general research program for testing when history and consequence have genuine control relevance without treating every organized dissipative process as an agent.

A second deterministic benchmark adds two harder adversaries to the original synthetic-controller benchmark. In one, a candidate history variable predicts action only because it proxies an unmeasured present variable; its gain collapses when present state is enriched. In the other, a nominal memory manipulation changes action through an off-target direct route; a selective memory probe has zero effect while an instrumented sham reproduces the naive intervention effect. A true writable-history controller survives both controls. These additional results validate the extended decision logic, not biological or general agency.

A third deterministic benchmark tests whether that logic survives five practical hazards: an intervention that reaches only part of its intended target, missing measurements and outcomes, rare positive outcomes, a changed test distribution, and dependence on one prediction family. The true writable-history construction retains positive history gain under all declared conditions. The incomplete-state proxy does not acquire positive out-of-distribution gain in either a regularized interaction-logic

model or a Laplace-smoothed fixed-bin model. Because the proxy sometimes harms shifted prediction, that result is reported as nonbenefit rather than falsely labeled equivalent to zero. The benchmark freezes construction-specific decision margins, multiplicity, power, and held-out-opening rules. It tests an analysis contract; it does not validate a biological substrate.

Keywords: teleonomy; agency; open systems; history-conditioned control; dissipation; causal intervention; information flow; mismatch reduction; active inference; bioelectricity

Scope and evidence boundary

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1. The problem: purpose language outruns measurement

Scientific descriptions often move too quickly from an observed outcome to an explanation in terms of purpose. A system returns to equilibrium, therefore it is said to seek equilibrium. A cell repairs a wound, therefore it is said to pursue a target morphology. A controller minimizes error, therefore it is said to possess a goal. A brain chooses an action, therefore a central chooser is presumed. These verbal compressions can be useful, but they also collapse important distinctions.

Four questions should be kept separate.

1. Does the system dissipate energy or produce entropy?
2. Does the system regulate a variable against disturbance?
3. Does stored history change its future control policy under matched present conditions?
4. Does the system satisfy operational criteria strong enough to justify conditional agency language?

The first question belongs to thermodynamics. The second belongs to control and cybernetics. The third concerns memory, learning, and causal history dependence. The fourth additionally concerns the organization of an individual, norms that matter to that individual, asymmetric control of coupling with an environment, and a repertoire of alternatives. A positive answer at one level does not automatically answer the later questions.

This paper reconstructs a teleonomic framework first developed within *Super Information Theory* (SIT) and turns it into a falsifiable control model. Its purpose is not to prove that nature has a universal aim. It is not a universal purpose field. It is not a new label for entropy production. It is not proof of consciousness. It is a proposal for identifying when a system's retained history and returned consequences have demonstrable control relevance.

2. Claim ceiling and contribution

The central claim is conditional:

Some open systems can be modeled as history-conditioned controllers. The teleonomic interpretation earns empirical support only when history improves held-out prediction or control relative to strong nulls, returned consequences causally update future policy, and additional agency criteria are independently measured.

The paper contributes five linked elements.

First, it defines a typed finite-horizon control objective whose variables must be operationalized for a particular substrate. Second, it defines a state-only comparator so that hidden state is not automatically renamed history. Third, it replaces a single agency score with a vector of separately interpretable claims. Fourth, it separates observational dependence from causal intervention and computational contraction from thermodynamic entropy production. Fifth, it supplies null models, experiments, and failure conditions that can defeat the proposal.

Draft 4 added an executable adversarial stress test in which incomplete present-state measurement and an uncontrolled direct intervention path each generate a false positive unless the corresponding control is applied. Draft 5 adds imperfect-intervention, missing-data, outcome-imbalance, distribution-shift, and model-family stress tests; freezes the synthetic decision rules; and supplies a preregistration contract for future substrate-specific studies.

The paper does not claim to invent cybernetics, teleonomy, autopoiesis, dissipative structures, active inference, transfer entropy, stochastic thermodynamics, or modern agency theory. Those are prior foundations [1-15]. Its narrower contribution is the joined operator:

perturbation

- > declared mismatch or viability pressure
- > internally accessible history and constraints
- > selection among available responses
- > physical action and dissipation
- > returned consequence
- > history update
- > altered response under a later matched condition.

Each arrow is an empirical obligation, not merely a narrative transition.

3. Genealogy and source boundary

The immediate first-party source is Chapter 51 of Micah Blumberg's SIT Version 78 manuscript, which bears an internal update date of December 18, 2025 [16]. The chapter's table-of-contents route is at source lines 401-427 and its relevant body spans lines 6553-7217 in the source-custody extract. That chapter joined entropy flux, a teleonomic potential, internal state, feedback, counterfactual policy structure, mismatch reduction, neural phase-wave differentials, biological development, and conditional agency. It also contained overextended language: a teleonomic force density was discussed before a frozen action and dimensional derivation existed; common coefficients were suggested across biology and gravitation; and a broad depolarization cycle was applied at multiple scales without substrate-specific identifiability.

Draft 2 preserves the useful research question and rejects those shortcuts. The SIT chapter is evidence of conceptual development, not external validation. Its local source custody is documented separately from this public-safe manuscript. Earlier SAN records provide broad ancestry in recurrent feedback, action-selection, choice, and changing future response, while the explicit joined teleonomic framework belongs to the later SIT stage. Mature 2025 terminology is not backdated into the 2011-2022 sources.

The ancestry can be stated more exactly without inflating it. A public Neo Mind Cycle video uploaded August 1, 2012 local time described a two-way EEG feedback conversation in which brain activity changed light and sound, the participant received the transformed consequence, and the next measured state changed. A second public August 23, 2012 account described a mirror-mediated closed loop from vision to neural and motor change to a changed image and renewed vision. The April 11, 2017 *Neural Lace Podcast* then asked whether conscious processing depends on feedback-connected traffic and proposed alternating connection-like and

connectionless transmission modes. Public 2022 SAN records added writable neural histories, recurrent route selection, sensory-action return, and explicit choice mechanisms. These are dated functional ancestors of the joined operator, not the mature teleonomic potential, claim vector, or agency test architecture. The 2025 SIT chapter is the first recovered first-party source that joins the entire package; this paper supplies its 2026 operationalization and adversarial tests.

The external genealogy is older and must be acknowledged. Ashby formalized regulation, variety, and control [1]. Wiener developed cybernetics around control and communication [2]. Biological teleonomy was developed to describe apparently goal-directed organization without making final causes a scientific primitive [3-5]. Maturana and Varela formulated autopoiesis as self-producing organization [6]. Beer provided dynamical and embodied routes to adaptive behavior [7]. Barandiaran, Di Paolo, and Rohde argued that agency requires individuality, interactional asymmetry, and normativity, and defined an agent in terms of autonomous organization that adaptively regulates environmental coupling in a self-sustaining way [8]. Friston and colleagues supplied influential free-energy and active-inference formalisms for perception, action, and learning [9,10]. Schreiber introduced transfer entropy as a directional dynamical dependence measure [11], while Ay and Polani separated observational dependence from interventional causal information flow [12]. Prigogine and Seifert provide distinct nonequilibrium and stochastic-thermodynamic foundations [13,14]. Levin provides a contemporary route from bioelectric pattern control to multiscale biological agency [15].

this paper therefore claims a synthesis and a test architecture, not priority over those component traditions.

4. Typed open-system model

Let $x_t \in \mathcal{X}$ be the current internal physical state of the modeled system, $e_t \in \mathcal{E}$ its measured environment, $h_t \in \mathcal{H}$ its internally accessible retained history, $u_t \in \mathcal{U}(x_t, e_t, h_t)$ an available control action, and $r_t \in \mathcal{R}$ a returned consequence. The distinction between x_t and h_t is functional rather than metaphysical. A variable belongs to h_t only when the system can access it and when intervention on that variable changes later control after present physical conditions have been matched as closely as the experiment permits.

The open-system dynamics are

$$dx_t = F(x_t, e_t, u_t; \theta) dt + G(x_t, e_t; \theta) dW_t. \quad (1)$$

Here F is a declared drift, G is a noise-amplitude map, W_t is a Wiener process when a diffusion approximation is appropriate, and θ contains fixed or learned parameters. Discrete systems can replace Equation (1) with a transition kernel.

History is updated by

$$h_{t+1} = \mathcal{U}_h(h_t, x_t, e_t, u_t, r_t; \eta), \quad (2)$$

where η contains learning, plasticity, developmental, or memory-update parameters. Equation (2) is not satisfied by an analyst's record that the system cannot use. The history must be physically implemented or otherwise causally available to the controller.

Let $\mathcal{V}_t \subseteq \mathcal{X}$ denote a declared viability set, target set, task-success region, or family of preferred states. A typed mismatch functional is

$$D_t = D(x_t, e_t, h_t; \mathcal{V}_t) \geq 0. \quad (3)$$

The meaning and units of D_t must be stated. It may be a squared tracking error, prediction loss, regret, deviation from a viability boundary, or a dimensionless divergence. Different choices cannot be combined without normalization and a substantive justification.

5. History-conditioned teleonomic potential

For a planning horizon T , define the effective control objective

$$J_H(\mathbf{u}_{t:t+T-1} \mid x_t, e_t, h_t) = \mathbb{E} \left[\sum_{s=t}^{t+T-1} (D_s + \lambda C_s + \beta q_s) + D_{t+T} \mid x_t, e_t, h_t \right]. \quad (4)$$

C_s is a declared action or constraint cost. The term q_s is an optional physical-resource accounting variable with explicit units; if it represents heat, work, or entropy production, that meaning must be established by a thermodynamic model. The coefficients λ and β carry whatever units are required to make the objective coherent. If those conversion choices are arbitrary, the analysis must report the separate objectives instead of claiming a natural scalar.

The history-conditioned policy is

$$\pi_H(x_t, e_t, h_t) \in \arg \min_{\mathbf{u}_{t:t+T-1}} J_H(\mathbf{u}_{t:t+T-1} \mid x_t, e_t, h_t). \quad (5)$$

Equations (4) and (5) define an effective teleonomic potential or landscape. They do not assert that the system consciously calculates a scalar. They say that a history-conditioned model can be tested against alternatives.

The primary comparator removes retained history while preserving present measured state, environment, model class, fitting budget, and control horizon:

$$\pi_0(x_t, e_t) \in \arg \min_{\mathbf{u}_{t:t+T-1}} J_0(\mathbf{u}_{t:t+T-1} \mid x_t, e_t). \quad (6)$$

This comparator is essential. Without it, a complicated present state can be called memory after the fact, and any successful response can be redescribed as history-conditioned.

The non-teleonomic limit is operational rather than rhetorical: if history does not improve held-out performance against the null family, or if intervention on the candidate history state does not change later policy, this experiment does not support history-conditioned teleonomic control.

6. A claim vector instead of a scalar agency index

Draft 1 combined conditional information quantities with an arbitrary null penalty in a single agency index. That construction mixed units, model choices, and causal statuses. Draft 2 replaces it with a claim vector whose components are evaluated independently.

Let $\mathcal{L}$ be a proper held-out predictive loss for future action or control outcome. The normalized history gain is

$$\Delta_H = \frac{\mathcal{L}(\pi_0) - \mathcal{L}(\pi_H)}{|\mathcal{L}(\pi_0)| + \varepsilon}, \quad \varepsilon > 0. \quad (7)$$

Positive Δ_H means that accessible history improves held-out prediction relative to the state-only comparator. It does not yet show that history is causal.

To test returned consequence, intervene on r_t while holding the intervention target and measured state fixed as closely as possible. Define

$$\Delta_R(r, r') = \frac{\|\mathbb{E}[h_{t+1} \mid do(r_t = r)] - \mathbb{E}[h_{t+1} \mid do(r_t = r')]\|}{s_h + \varepsilon}, \quad (8)$$

where s_h is a declared scale for the history variable. A corresponding downstream policy effect is

$$\Delta_\pi(r, r') = d_\pi\left(\pi_{t+1}^{do(r_t=r)}, \pi_{t+1}^{do(r_t=r')}\right), \quad (9)$$

with d_π a predeclared policy distance. Equations (8) and (9) test the chain consequence $\rightarrow$ history $\rightarrow$ later policy.

Let $\mathcal{U}_{cf}(x, e, h)$ be the set of actions that remain feasible under declared counterfactual perturbations. A normalized repertoire statistic is

$$K_{cf} = \frac{\log(1 + |\mathcal{U}_{cf}(x, e, h)|)}{\log(1 + |\mathcal{U}_{max}|)}. \quad (10)$$

This is only a coarse repertoire measure. For continuous action spaces, the action set must first be partitioned into preregistered, experimentally distinguishable outcome classes. Stronger analyses can use empowerment or intervention-conditioned reachable sets, but raw action count must not be confused with meaningful alternatives.

The experimental claim vector is

$$\mathbf{Z}_T = (\Delta_H, \Delta_R, \Delta_\pi, K_{cf}, C_{ind}, C_{norm}, C_{asym}). \quad (11)$$

The final three entries are criterion-specific estimates with uncertainty:

- C_{ind} : evidence that the candidate system has a defensible operational boundary and maintains its organization over the evaluation interval;
- C_{norm} : evidence that outcomes can be better or worse relative to the system's own viability or maintained organization, rather than only relative to the experimenter's preference;
- C_{asym} : evidence that the candidate system regulates its coupling with the environment, rather than merely being driven by it.

No weighted total is reported. A system can have large history gain and weak individuality, or strong regulation and no demonstrated counterfactual repertoire. The vector preserves those differences. It is not proof of consciousness, subjective experience, moral status, or legal personhood.

7. Observation routes candidates; intervention tests causality

Conditional mutual information can identify dependence between history and later action:

$$I(H_t; U_{t+1} \mid X_t, E_t) = \sum p(h, u, x, e) \log \frac{p(u \mid h, x, e)}{p(u \mid x, e)}. \quad (12)$$

Transfer entropy provides a directional dynamical dependence measure by comparing prediction of a target process with and without the source history [11]. These quantities are useful routing statistics. They do not by themselves rule out latent common causes, measurement leakage, model misspecification, polyadic dependence, or a present state that incompletely captures the system [31].

The causal estimand must instead be tied to an intervention. For a later action feature Y , define

$$\text{ACE}_{h \rightarrow Y}(h, h') = \mathbb{E}[Y \mid \text{do}(H_t = h)] - \mathbb{E}[Y \mid \text{do}(H_t = h')]. \quad (13)$$

The intervention may be physical, computational, pharmacological, optogenetic, bioelectric, lesion-based, or implemented by a validated natural experiment. The manipulation must not change the outcome through an uncontrolled direct route. Ay and Polani’s distinction between observational and interventional conditioning is decisive here [12]. An observational transfer measure can nominate a channel; an intervention is needed to argue that the channel controls the outcome.

8. Computational mismatch is not thermodynamic entropy

The word “dissipation” is used in several literatures. At least three meanings must be separated.

1. A computational loss decreases during an update.
2. A Lyapunov or mismatch functional contracts along a trajectory.
3. A physically defined entropy production is nonnegative on average under specified thermodynamic conditions.

These meanings can be related in a concrete model, but they are not synonyms.

For a declared mismatch D_t , write the local balance as

$$\frac{dD_t}{dt} = -\mathcal{R}_t + \mathcal{P}_t, \quad \mathcal{R}_t \geq 0. \quad (14)$$

$\mathcal{R}_t$ is the modeled contraction or removal rate, and $\mathcal{P}_t$ is mismatch production from forcing, changing targets, unresolved variables, or disturbances. Equation (14) is a bookkeeping certificate. Contractive behavior requires an interval on which $\mathcal{R}_t > \mathcal{P}_t$. This is the bounded bridge to Micah’s New Law of Thermodynamics [17]. It does not identify physical entropy unless D , $\mathcal{R}$, and $\mathcal{P}$ have been linked to a thermodynamic model.

If the action coordinate follows a true gradient flow with a positive-semidefinite mobility matrix K ,

$$\dot{u}_t = -K \nabla_u \Phi_T(x_t, e_t, h_t, u_t), \quad (15)$$

then the total derivative is

$$\frac{d\Phi_T}{dt} = -\nabla_u \Phi_T^\top K \nabla_u \Phi_T + \nabla_x \Phi_T^\top \dot{x}_t + \nabla_e \Phi_T^\top \dot{e}_t + \nabla_h \Phi_T^\top \dot{h}_t + \partial_t \Phi_T. \quad (16)$$

Only the first term is guaranteed nonpositive. Draft 1 incorrectly assigned that gradient term directly to (dD/dt) . Equation (16) shows the missing terms and the conditions needed for a monotonic-potential claim.

For a thermodynamically resolved stochastic trajectory ω and its appropriately reversed trajectory $\tilde{\omega}$, define dimensionless total entropy production by the path-probability ratio

$$\Sigma[\omega] = \ln \frac{P_F[\omega]}{P_R[\tilde{\omega}]}.$$
 (17)

Under the conditions of the integral fluctuation theorem,

$$\begin{aligned} \langle e^{-\Sigma} \rangle &= 1, \\ \langle \Sigma \rangle &\geq 0. \end{aligned}$$
 (18)

Seifert’s trajectory-level framework supplies this thermodynamic route [14]. Entropy production does not establish agency. A passive driven system can produce entropy. Conversely, a controller can be described computationally before its complete energetic accounting is known. The scientific bridge is earned only when the computational states, physical microstates or mesostates, reservoirs, time-reversal convention, and measurement boundary are specified.

9. The obligation ladder

Draft 1 used a conceptual set-inclusion chain in which contractive mismatch dynamics were placed inside history-conditioned control and history-conditioned control inside agency. That was logically wrong. Many contractive systems have no writable history, and many history-conditioned controllers do not satisfy a defensible agency criterion.

Draft 2 replaces inclusion with an obligation ladder.

Level	Claim	Minimum evidence	What the level does not establish
0	Open-system dissipation	Energy or entropy accounting with boundaries and reservoirs	Regulation, memory, purpose, or agency
1	Regulation	Maintained variable, disturbance, feedback path, and matched open-loop control	Learning or history-conditioned policy
2	History-conditioned adaptive control	Positive held-out history gain plus consequence-to-history-to-policy intervention chain	Individuality, system-relative normativity, rich alternatives, or consciousness
3	Conditional agency	Level 2 plus individuality, normativity, interactional asymmetry, and a counterfactual repertoire	Subjective experience, moral status, or cosmic purpose

The levels are evidential packages. They are not claims that every instance at one level is literally a mathematical subset of every instance at the next.

10. Required null models

A history-conditioned model is easy to overfit. The following null family is mandatory.

State-only null. Remove h_t while preserving the measured present state, environment, horizon, optimizer, and data budget.

Time-shuffled-history null. Permute historical sequences within matched contexts. This tests whether temporal order matters rather than merely extra covariates.

Consequence-permuted null. Break the pairing between action consequences and later history updates. This tests whether the returned consequence carries specific learning relevance.

Capacity-matched recurrent null. Compare against a model with equal parameter count and recurrence but no privileged teleonomic structure. This tests whether improvement is generic capacity.

Energy-matched null. For physical systems, match energy input, temperature, duration, actuator use, and measurement burden. This prevents a more energetic controller from being declared more agentic merely because it has a larger intervention budget.

Post-hoc narrative null. Freeze the target, viability measure, history variables, and success criterion before the held-out evaluation. This tests whether the apparent goal was chosen after the observed trajectory.

Observer-only-memory null. Supply the analyst with history while preventing the system from accessing it. If prediction improves but intervention on the candidate system memory does not change policy, the result supports a better observer model rather than system-level history-conditioned control.

Present-state-enrichment null. Before treating a candidate variable as retained history, add plausible currently active variables, higher-resolution measurements, and matched nuisance channels to the present-state comparator. If the apparent history gain collapses within a preregistered equivalence region, the result supports incomplete present-state measurement rather than a distinct memory contribution.

Instrumented direct-path sham. Reproduce the physical burden, timing, spread, and off-target consequences of a nominal memory manipulation while holding the candidate history variable at a neutral or rescued value. If the sham reproduces the action effect, the original manipulation does not isolate the memory-to-policy route.

Passing one null is insufficient. The null family should be preregistered and adjusted to the substrate.

11. Experimental hypotheses

Four core hypotheses can be tested across synthetic and biological systems.

H1: history relevance. Under held-out matched conditions, $\Delta_H > 0$ against the state-only, shuffled-history, and capacity-matched nulls.

H2: consequence writing. Intervention on returned consequence changes the candidate history variable, producing $\Delta_R > 0$ with a confidence interval excluding the preregistered equivalence region.

H3: memory-to-policy control. Selective intervention on the history variable changes later policy while preserving current sensory input and major nuisance variables, producing a nonzero $ACE_{h \rightarrow Y}$.

The word “selective” is an experimental burden. A nominal memory intervention must not count as H3 when its behavioral effect is reproduced by an instrumented direct-path sham or disappears when the off-target route is held neutral.

H4: alternatives. The system maintains more than one reachable response under a controlled counterfactual manipulation, and the selected response changes in the predicted direction when the represented alternative or norm is manipulated.

These hypotheses form a causal chain. H1 without H2 and H3 may reflect observational leakage. H2 without H3 may show storage without control. H3 without H4 may show a learned reflex. H4 without individuality and normativity may show a flexible optimizer without a strong agency interpretation.

12. Test program A: synthetic controllers

The cleanest first test uses simulated or robotic systems because complete states, interventions, and energetic proxies can be controlled.

Construct four controllers with the same sensors and actuators:

1. an open-loop controller;
2. a memoryless feedback controller;
3. a recurrent learned controller with writable history; and
4. a planner with an explicit counterfactual repertoire.

Expose each controller to training environments and then to held-out perturbations in which current observations are matched but relevant prior histories differ. Fit π_0 and π_H using identical capacity and compute budgets. Perform targeted memory swaps, consequence substitutions, and policy rescue interventions. Measure task outcome, control cost, energy use, $\mathbf{Z}_T$, and all null-model results.

The framework predicts that the memoryless feedback controller can pass Level 1 while failing H1-H3. The recurrent controller may pass Level 2. The planner can qualify for Level 3 only if its counterfactual repertoire is causally used and the individuality, normativity, and asymmetry criteria are operationalized rather than assumed from software labels.

12A. Reference benchmark: predictive history is not yet system memory

The first executable benchmark instantiates the distinction that motivates the paper. Five synthetic systems receive the same present-state variable and binary action interface:

1. an open-loop controller;
2. a memoryless feedback controller;
3. a controller whose writable internal history changes later action;
4. a planner whose history also changes its counterfactual action repertoire; and
5. an observer-only control in which an analyst’s record predicts later action but cannot be accessed or changed through the candidate system memory.

For every system and every seed, the benchmark fits four-parameter logistic comparators with the same optimization and regularization procedure. The state-only model receives present state plus matched nuisance features. The history model replaces the nuisance channel with the candidate history and its interaction with present state. Time-shuffled and capacity-matched controls use the same feature count. Forty-eight fixed seeds per system are evaluated on held-out events. Separate interventions set returned consequence and system history, so predictive gain cannot substitute for Equations (8), (9), and (13).

Synthetic system	Median Δ_H	Median Δ_R	Median memory-to-policy	
			ACE	H1-H4 disposition
Open loop	0.0002	0.0000	0.0000	Fails H1-H4; Level 0 control
Memoryless feedback	0.0002	0.0000	0.0000	Fails H1-H4; Level 1 regulation only
Writable-history controller	0.3546	1.2130	0.5763	Passes H1-H3; Level 2 diagnostic package
Counterfactual planner	0.4539	1.2130	0.6613	Passes H1-H4; Level 3 diagnostic package only

Synthetic system	Median Δ_H	Median Δ_R	Median memory-to-policy	
			ACE	H1-H4 disposition
Observer-only history	0.3579	0.0000	0.0000	Predictive history, but fails the causal chain

For the writable-history controller, the fifth percentile of held-out history gain is 0.3325; for the planner it is 0.4216. The ninety-fifth-percentile gains of the time-shuffled comparator remain below 0.002 for both systems, and the capacity-matched nuisance comparator has no positive gain in the frozen reference construction. The observer-only record is deliberately adversarial: its fifth-percentile predictive gain is 0.3393, yet returned-consequence and candidate-memory interventions have zero effect. It therefore defeats any rule that promotes predictive history alone to system-level history-conditioned control.

The benchmark is a smoke test for the paper’s logic. The controller classes, data-generating equations, thresholds, and dispositions were chosen to make the expected distinctions identifiable. Passing them shows that the proposed ladder can keep dissipation, feedback, history-conditioned control, and the stronger agency package apart under declared conditions. It does not estimate the prevalence of agency, validate a biological history variable, establish consciousness or moral status, or show that the synthetic planner is an agent outside the benchmark’s stipulated diagnostic criteria. A serious empirical study must freeze its own controller family, state completeness tests, nuisance measurements, interventions, equivalence region, power analysis, and held-out seal before evaluation.

12B. Why a record can look causal when the present is incompletely measured

Consider two controllers that show the same value on a dashboard. One controller also has a hot battery, but temperature was not measured. An archived warning flag happens to correlate with battery temperature. The warning flag can improve prediction of the controller’s next action even if the controller never reads the archive. Once temperature is measured, the archive may add nothing. The first analysis did not discover memory control; it discovered an incomplete description of the present.

This is the ordinary problem that must be solved before a new diagnostic name is useful. A candidate history variable can proxy a currently active molecular, electrical, mechanical, computational, or environmental state omitted from x_t . Matching only the recorded present does not match the actual present. A strong test therefore builds a sequence of increasingly complete present-state descriptions and asks whether history retains incremental predictive value.

Let $X_t^{(0)}$ be the initially measured present state and let $X_t^{(1)} = (X_t^{(0)}, Z_t)$ include a preregistered additional present-state variable. For either measurement level k , define

$$\Delta_H^{(k)} = \frac{\mathcal{L}(\pi_0^{(k)}) - \mathcal{L}(\pi_H^{(k)})}{|\mathcal{L}(\pi_0^{(k)})| + \varepsilon}, \quad (19)$$

$$k \in \{0, 1\}.$$

Only after this operation is clear do we name it the **present-state-enrichment test**. Its diagnostic contrast is

$$\mathcal{C}_{state} = \Delta_H^{(0)} - \Delta_H^{(1)}. \quad (20)$$

A large $\mathcal{C}_{state}$ is not positive evidence for history. If $\Delta_H^{(1)}$ enters the preregistered equivalence region while a capacity-matched enriched-state model performs equally well, the apparent history result has been explained by missing present-state information. If history gain survives successive plausible state enrichments and a selective history intervention, confidence in a distinct history route increases. No finite sensor set proves that the state is complete; the test is a controlled sensitivity analysis, not an omniscience claim.

12C. Why changing a proposed memory can change action for the wrong reason

Now consider an electrode, drug, software patch, or field manipulation intended to change a memory variable. The manipulation also excites a motor pathway, changes arousal, alters sensory input, or consumes a different amount of energy. Behavior changes after the intervention. That total effect does not yet show that memory caused the change, because the manipulation had another route to the outcome.

The established causal problem is an exclusion failure or uncontrolled direct path [12,26]. Let M_t be the manipulation assignment, H_t the candidate history variable, B_t the measured or modeled bypass route, and Y_{t+1} the later action feature. A minimal structural description is

$$H_t = g(X_t, M_t, \epsilon_H), \quad B_t = b(M_t, \epsilon_B), \quad Y_{t+1} = f(X_t, H_t, B_t, \epsilon_Y). \quad (21)$$

The naive total effect of the manipulation is

$$\text{TE}_M(m, m') = \mathbb{E}[Y_{t+1} \mid \text{do}(M_t = m)] - \mathbb{E}[Y_{t+1} \mid \text{do}(M_t = m')]. \quad (22)$$

Equation (22) mixes every path created by M_t . The memory claim requires a more selective contrast that changes H_t while holding the declared bypass route at a neutral value b_0 :

$$\text{ACE}_{H \rightarrow Y}^{\text{sel}}(h, h'; b_0) = \mathbb{E}[Y_{t+1} \mid \text{do}(H_t = h), \text{do}(B_t = b_0)] - \mathbb{E}[Y_{t+1} \mid \text{do}(H_t = h'), \text{do}(B_t = b_0)]. \quad (23)$$

The complementary control changes the bypass route while holding candidate history neutral:

$$\text{ACE}_{B \rightarrow Y}^{\text{sham}}(b, b'; h_0) = \mathbb{E}[Y_{t+1} \mid \text{do}(B_t = b), \text{do}(H_t = h_0)] - \mathbb{E}[Y_{t+1} \mid \text{do}(B_t = b'), \text{do}(H_t = h_0)]. \quad (24)$$

Only after the contrast is understood do we call Equation (24) an **instrumented direct-path sham**. A nominal memory effect fails the selective test when Equation (23) is equivalent to zero and Equation (24) reproduces Equation (22). In a physical experiment, B_t will rarely be a single measured variable. The burden is to measure likely spread, timing, arousal, energy, sensory, actuator, and lesion routes; use sham and rescue conditions; and state which exclusion assumptions remain unverified.

12D. Adversarial confound stress test

The Draft 4 extension instantiates the two problems before promoting their names. It uses 48 fixed seeds for each of three synthetic systems:

1. a true writable-history controller whose candidate history causally changes action;
2. an incomplete-present-state system in which the candidate “history” is only a noisy proxy for an omitted currently active variable; and
3. an uncontrolled-direct-path system in which a nominal memory manipulation reaches action only through a bypass route.

Every predictive comparison uses the same regularized logistic fitting procedure and held-out loss. The base comparison omits one present-state variable. The enriched comparison adds that variable and a capacity-matched nuisance channel. Three intervention probes are then kept separate: the naive manipulation changes both candidate memory and the possible bypass, the selective probe changes memory while holding the bypass neutral, and the instrumented sham changes the bypass while holding memory neutral.

Synthetic system	Median base history gain	Median residual history gain after state enrichment	Median naive effect	Median selective memory effect	Median direct-path sham
True writable history	0.3750	0.3752	0.6070	0.6070	0.0000
Incomplete present state	0.2533	-0.0018	0.0000	0.0000	0.0000
Uncontrolled direct path	0.0002	-0.0001	0.6134	0.0000	0.6134

All seven frozen stress-test checks pass. In the incomplete-state adversary, the fifth percentile of base history gain is 0.2313, yet the ninety-fifth percentile of residual gain after state enrichment is 0.0013. In the direct-path adversary, the fifth percentile of the naive effect is 0.6107, the selective memory effect is exactly zero in the declared construction, and the sham reproduces the naive effect. The true-history controller retains a fifth-percentile residual history gain of 0.3561 and a fifth-percentile selective intervention effect of 0.6033 while its direct-path sham remains zero.

These are deliberately identifiable synthetic cases. They show that the framework can reject two false positives that Draft 3 only named as future work. They do not show that any empirical present-state measurement is complete, that any real intervention is perfectly selective, or that the resulting controller is a biological or general agent. Those questions require substrate-specific measurement, negative controls, rescue, sensitivity analysis, and independent review.

12E. Why one clean benchmark is not enough

The earlier benchmarks assume complete observations, approximately balanced outcomes, one stable data distribution, one prediction family, and a manipulation that exactly reaches its intended state. Those assumptions make the logic visible, but they are not a sufficient analysis contract. A real memory perturbation may reach only part of its target. Some candidate histories or outcomes may be unobserved. Successful actions may be rare. The test population may differ from the development population. A flexible predictor can also make a result appear to belong to the system when it belongs to the analyst’s chosen model.

Those problems should be understood before assigning technical names to them. If a manipulation is intended to move history from h_0 to h_1 but reaches only a fraction γ of that change, the selective contrast becomes

$$\begin{aligned}
 h^{(\gamma)} &= h_0 + \gamma(h_1 - h_0), \\
 \tau_H(\gamma; b_0) &= \mathbb{E}[Y \mid do(H = h^{(\gamma)}), do(B = b_0)] - \mathbb{E}[Y \mid do(H = h_0), do(B = b_0)].
 \end{aligned}
 \tag{25}$$

Only after the partial-target problem is explicit do we call γ **intervention fidelity**. Fidelity must be estimated from target-engagement measurements rather than inferred from the behavioral outcome. A monotonic change in τ_H as target engagement increases is useful evidence, but it cannot rule out every correlated off-target route. The selective contrast, instrumented sham, and rescue remain necessary.

If an observation is missing, silently dropping it can change both the population and the comparison. Let $R_i = 1$ when the relevant training outcome is observed, let $p_i = P(R_i = 1 \mid Z_i)$ be a preregistered or estimated observation probability based on measured variables Z_i , and let ℓ_i be the declared loss. One correction is

$$\hat{\mathcal{L}}_{\text{IPW}} = \frac{\sum_i R_i \hat{p}_i^{-1} \ell(y_i, \hat{y}_i)}{\sum_i R_i \hat{p}_i^{-1}}. \quad (26)$$

This is an inverse-probability-weighted loss. It is justified only under the declared observation model, positivity, and adequate measurement of the variables that govern missingness. It does not repair data that are missing for an unmeasured reason. The analysis must also include explicit missingness indicators, report observation fractions by class and condition, and compare complete case, weighted, and sensitivity analyses.

When one outcome class is rare, ordinary average loss can reward a model that mainly predicts the majority. With n_1 positive and n_0 negative held-out cases, the class-balanced log loss is

$$\mathcal{L}_{bal} = \frac{1}{2} \left[-\frac{1}{n_1} \sum_{i:y_i=1} \log \hat{p}_i - \frac{1}{n_0} \sum_{i:y_i=0} \log(1 - \hat{p}_i) \right]. \quad (27)$$

This earns the label **class-imbalance control**. It does not make prevalence irrelevant. The paper must report prevalence, class-conditional performance, calibration, and the consequence of false positive and false negative decisions.

A controller may appear history-dependent on one familiar population and fail after the state distribution changes. Let $\mathcal{Q}$ be a small, frozen family of scientifically plausible test distributions. The worst-case history gain is

$$\Delta_H^{rob} = \min_{Q \in \mathcal{Q}} \frac{\mathcal{L}_Q(\pi_0) - \mathcal{L}_Q(\pi_H)}{|\mathcal{L}_Q(\pi_0)| + \varepsilon}. \quad (28)$$

Only now do we call this a **distribution-shift stress test**. The set $\mathcal{Q}$ must be chosen before the confirmatory data are opened. An unbounded claim of robustness cannot be inferred from a finite set of shifts.

Finally, let $\mathcal{M}$ be a frozen family of prediction methods that embody meaningfully different approximation assumptions. Define

$$\begin{aligned} \Delta_H^{min} &= \min_{m \in \mathcal{M}} \Delta_H^{(m)}, \\ S_{\mathcal{M}} &= \max_{m \in \mathcal{M}} \Delta_H^{(m)} - \min_{m \in \mathcal{M}} \Delta_H^{(m)}. \end{aligned} \quad (29)$$

Δ_H^{min} asks whether positive history gain survives every declared family; $S_{\mathcal{M}}$ reports model-family sensitivity. Agreement between two families is not a proof of model independence. Disagreement is a reason to narrow the claim or investigate misspecification, not to select the favorable model after seeing the held-out result.

12F. Robustness benchmark and results

The Draft 5 benchmark uses 48 fixed seeds per condition and retains the typed systems from Draft 4. The true-history construction makes a writable binary history state control later action. The incomplete-state adversary

makes the candidate history a noisy proxy for an active state variable that is available to the enriched state-only comparator. The direct-path construction routes a naive manipulation around the candidate history. A mixed construction contains both a genuine history path and a bypass.

The missing-data test independently masks candidate measurements and training outcomes. The first condition observes each with probability 0.75. In the second, observation probability changes with the magnitude of measured present state. The synthetic propensity is known by construction, so Equation (26) is used for observed training outcomes and a missingness indicator is supplied for candidate measurements. This is a decision-logic test, not evidence that the same correction will identify a real missing-not-at-random process.

The imbalance condition shifts the logit intercept so that the median held-out positive prevalence is 0.1786 and evaluates Equation (27). The shift condition moves and widens the present-state distribution and increases proxy corruption. Two deliberately different families are fitted: a regularized logistic model with declared interactions and a Laplace-smoothed fixed-bin conditional-probability model.

Stress condition	True or mixed-history result across 48 seeds	Incomplete-state or direct-path control	Frozen interpretation
25 percent intervention fidelity	True selective effect median 0.1734; fifth percentile 0.1719. Mixed selective effect median 0.1161; fifth percentile 0.1151.	Direct-path selective effect is exactly zero; direct-path sham median 0.3378 reproduces its naive effect.	Partial target engagement weakens but does not erase the constructed history path; the bypass still fails the selective contrast.
25 percent completely random missingness	True median history gain 0.2822; fifth percentile 0.2666.	Proxy gain fifth to ninety-fifth percentile: -0.0050 to 0.0033.	True gain survives; proxy is equivalent to zero inside the construction-specific plus or minus 0.02 region.
Present-dependent missingness	True median history gain 0.2723; fifth percentile 0.2512.	Proxy gain fifth to ninety-fifth percentile: -0.0041 to 0.0026.	True gain survives the declared weighted analysis; proxy remains equivalent to zero.
Rare positive outcomes	True balanced gain median 0.3869; fifth percentile 0.3636.	Proxy balanced gain fifth to ninety-fifth percentile: -0.0095 to 0.0050.	The result is not a majority-class artifact in this construction.
Shifted distribution, logistic family	True median gain 0.4364; fifth percentile 0.4211.	Proxy median gain -0.0196; ninety-fifth percentile -0.0106.	True gain survives; proxy has no positive gain and sometimes harms shifted prediction. It is not called equivalent to zero.
Shifted distribution, fixed-bin family	True median gain 0.4214; fifth percentile 0.4028.	Proxy median gain -0.0100; ninety-fifth percentile -0.0033.	The positive result survives a second family; the proxy again has no positive gain.

All twelve declared checks pass. One implementation lesson is scientifically important. An early version of the decision code inspected only the upper tail while describing a two-sided equivalence result. Correcting the code exposed that the shifted logistic proxy's lower tail fell outside the equivalence band. Draft 5 therefore makes the narrower and accurate claim: the proxy does not improve shifted prediction across either model family; it is not asserted to be equivalent to zero in that condition. The executable result records the corrected rule and hashes the frozen decision file.

These results close the five local synthetic stress-test gaps. They do not establish intervention fidelity, ignorable missingness, transportability, adequate power, or model completeness in an empirical system. Those properties must be measured and challenged in the substrate-specific study.

The pattern across benchmarks is more informative than a single pass label. The true-history construction is a sensitivity control: a defensible procedure should retain its effect when the declared nuisance conditions are introduced. The observer-only, incomplete-state, and direct-path constructions are specificity controls: a defensible procedure should not promote them merely because they predict, correlate, or respond to a nominal intervention. The mixed construction shows why a system need not be classified all-or-nothing. A real history route can coexist with a bypass, so the selective effect, bypass effect, and total effect must all be reported. The bypass does not erase the genuine route, and the genuine route does not excuse the bypass.

No total robustness score is formed. Missingness, imbalance, shift, target engagement, and model-family dependence answer different questions and can fail independently. A study that survives four hazards but fails the fifth should identify the failed obligation rather than average it away. This mirrors the paper's treatment of agency: evidence is carried as a vector of claims and limitations, not converted into one impressive number.

12G. Frozen decision rules and held-out seal

A finite benchmark becomes easy to overfit when thresholds, endpoints, or model families change after the results are visible. Draft 5 therefore freezes the following rules for its synthetic claims and carries them into a substrate-specific preregistration template.

- The confirmatory claim family is H1 through H4 plus any declared substrate-specific co-primary endpoint. Holm's step-down procedure controls family-wise error at 0.05. Equivalence uses two one-sided tests at the multiplicity-adjusted level.
- For this synthetic construction only, normalized history gain is equivalent to zero inside plus or minus 0.02, and a probability-difference effect is equivalent to zero inside plus or minus 0.05. These are not universal biological effect sizes.
- A positive synthetic claim requires the fifth percentile across the 48 fixed seeds to exceed its declared threshold. A two-sided equivalence claim requires both the fifth and ninety-fifth percentiles inside the equivalence region. A nonbenefit claim requires only that the ninety-fifth percentile not exceed the positive edge; negative values are not mislabeled equivalent to zero.
- The declared fifth-percentile thresholds are 0.05 for the minimum-fidelity selective probability difference, 0.10 for completely random missingness, 0.08 for present-dependent missingness, 0.10 for class-balanced history gain, 0.10 for the shifted logistic family, and 0.06 for the shifted fixed-bin family.
- A future empirical study targets 90 percent simulation-based power at the smallest effect outside a prospectively justified equivalence region, including the worst credible attrition or missingness condition. An underpowered result is inconclusive unless a prospectively powered equivalence test passes. Optional stopping is prohibited.
- Before the confirmatory held-out set is opened, the target and viability definition, boundary, history and present-state variables, model and null families, interventions and shams, equivalence regions, power simulation, multiplicity family, missing-data rule, shift set, code, and environment hashes must be sealed.
- The confirmatory set is opened once. Any refit, threshold change, endpoint change, or model-selection decision after opening creates an exploratory version and requires a fresh untouched confirmatory set for renewed confirmation.

These rules are frozen for Draft 5's declared benchmark. Substrate-specific margins still require prospective scientific and practical justification; they cannot be copied from the synthetic construction merely because the code passes.

13. Test program B: morphogenesis and bioelectric control

Developmental and regenerative systems are valuable because tissues can reach similar anatomical outcomes through different paths after damage. Levin’s bioelectric-network work supplies a relevant comparator for multiscale pattern memory and goal-directed language [15]. The present framework asks a narrower causal question: does a physically identifiable retained pattern change the repair policy when local present conditions are matched?

A preregistered experiment would specify:

- a target morphology and quantitative distance metric;
- the candidate bioelectric, transcriptional, or structural history variable;
- a perturbation that changes returned consequence without simply changing tissue damage;
- a memory intervention, such as a bounded bioelectric-state manipulation;
- sham, energy-matched, state-only, and capacity-matched mechanistic models;
- a rescue that restores the candidate history variable without restoring every upstream condition; and
- blinded anatomical and functional endpoints.

Positive evidence would require the consequence intervention to update the candidate state and the selective state intervention to alter the later repair route as predicted. Regeneration alone is insufficient. A stereotyped biochemical cascade can restore morphology without satisfying the Level 3 criteria.

14. Test program C: neural control and choice

SAN, the Phase Wave Differential framework, and the Choice paper propose a neural implementation route in which local differences in timing, recruitment, and electrical drive alter which downstream pathway gains control [18-20]. *Coherence-Weighted Electrical Drive in Choice and Motor Realization* adds a candidate bridge from coherence-weighted electrical drive to motor realization and sensory return [21]. this paper does not require that implementation. It supplies the more general history-and-consequence obligations that any neural mechanism must satisfy.

An experiment can compare present-state and history-conditioned models of a choice task while perturbing a candidate memory trace, outcome signal, or route-selection circuit. It should separate:

- stimulus history from internally retained history;
- prediction of choice from causal control of choice;
- oscillatory correlation from pathway-specific intervention;
- motor preparation from conscious report; and
- adaptive policy from a fixed stimulus-response mapping.

The minimum result is not a decoding accuracy increase. It is a selective causal chain in which an earlier returned consequence changes a retained state, intervention on that state changes later route selection, and a matched rescue restores the predicted behavior.

15. Relation to adjacent Micah Blumberg papers

Paper 75 is designed to close a gap rather than duplicate the existing corpus.

Adjacent work	Existing role	Paper 75 addition
<i>A Mathematical Framework for Contractive Computational Dynamics with a Thermodynamic Bridge Super Information Theory</i>	General mismatch balance and contraction certificates	Adds writable history, returned consequence, strong nulls, and agency obligations
<i>Self Aware Networks</i> and PWD paper	Parent architecture linking information, time, dynamics, and cross-domain hypotheses Neural receiving, transformation, projection, oscillatory difference, and learning mechanisms	Extracts and repairs the removed teleonomic / Telos layer as a bounded companion program Provides substrate-general control criteria without requiring the SAN mechanism
<i>Choice Without a Central Chooser</i>	Distributed neural route selection	Distinguishes route selection from broader conditional agency
Paper 74 on coherence-weighted electrical drive	Candidate neural chain from recruitment to motor output and sensory return	Generalizes the history-consequence-control structure without universal beta / gamma claims
TCNR	Consciousness-specific indicator and perturbation program	Supplies a non-consciousness-specific agency vector; it does not collapse agency into consciousness
Super Evolution Theory	Evolutionary and multigenerational search	Focuses on within-episode, developmental, or learned history-conditioned control
Restored God Proof	Metaphysical and modal argument using selected scientific premises	Clarifies one possible agency premise but does not by itself establish a necessary mind

The distinction from TCNR is especially important. A consciousness indicator battery cannot be reduced to the agency vector, and the agency vector cannot be advertised as consciousness evidence without the additional TCNR burdens.

16. Active inference and other explanatory competitors

Active inference is a serious comparator, not a vocabulary donor to be renamed. It models perception and action through variational free energy, generative models, and policy-related quantities [9,10]. Paper 75 differs in emphasis: it begins with a model-agnostic history intervention and an evidential ladder. A given active-inference model may instantiate the framework, but the framework does not assume that every controller implements variational inference.

Likewise, autopoiesis emphasizes self-producing organization [6], dynamical-systems approaches emphasize coupled agent-environment trajectories [7], and modern agency theory emphasizes individuality, normativity, and asymmetry [8]. This paper adopts these as constraints on strong agency language. It does not replace them with a new coefficient.

The decisive comparison question is operational: which framework makes a distinct prediction under a history swap, consequence intervention, memory lesion, matched rescue, or counterfactual alternative manipulation? If two theories predict the same interventions, the paper should report functional equivalence at that tested resolution rather than claim a unique mechanism.

17. Buoyancy, water, ice, and phase-change boundary

Earlier SIT development used buoyancy and water-state changes as intuition for density, coherence, phase, and gravitational response. That material remains useful as a source of questions, but ordinary buoyancy

is already explained by standard pressure gradients, displaced fluid, density, geometry, temperature, and stress-energy accounting. Freezing and vaporization change density, bonding, latent heat, boundary forces, convection, and apparatus behavior. Those ordinary changes must be modeled before any residual interpretation.

For a preregistered residual experiment, define

$$\Delta a_{res} = a_{measured} - a_{standard}(\rho, p, T, V, \Gamma, \sigma, E_{total}), \quad (30)$$

where Γ summarizes geometry and boundary conditions and σ summarizes relevant stresses. Equation (30) preserves the Draft 3 residual-test definition while keeping Draft 5's equations sequential. A proposed SIT-related coupling can be tested only after the standard model and uncertainty budget are frozen. Phase change is not evidence for such a coupling merely because density and microscopic coherence change.

This appendix-level route does not support a separate paper at present. It also does not support a claim that ordinary buoyancy contradicts existing physics.

18. Failure conditions and adversarial checks

The framework must be narrowed or rejected for a substrate if any of the following holds.

1. History-conditioned models do not outperform properly matched state-only and capacity-matched nulls.
2. The apparent history gain disappears when current state is measured more completely.
3. Time-shuffled or consequence-permuted histories perform equally well.
4. Intervention on returned consequence does not change the candidate history variable.
5. Intervention on the candidate history variable does not change later policy.
6. The intervention changes policy through an uncontrolled direct path.
7. The proposed norm is imposed only by the experimenter and has no relation to maintained system organization or viability.
8. The candidate boundary is arbitrary and the individuality result is not robust to nearby boundaries.
9. Interactional asymmetry disappears under force, energy, or actuator matching.
10. The counterfactual repertoire is only a large action alphabet with no demonstrated use of alternatives.
11. The thermodynamic terms lack defined reservoirs, units, or path probabilities.
12. A decrease in computational loss is presented as physical entropy production without a bridge model.
13. The target or success criterion was selected after observing the trajectory.
14. A simpler feedback or dynamical model predicts every intervention equally well.
15. Domain-specific variables are hidden under shared symbols even though their units and causal roles differ.
16. The effect disappears at realistic target-engagement levels or does not vary with measured intervention fidelity as predicted.
17. Missingness sensitivity includes zero or reverses the result under a credible observation model.
18. The result is explained by outcome prevalence, poor minority-class calibration, or a majority-only decision rule.
19. Positive history gain fails on a prospectively declared shifted distribution.
20. The conclusion changes across reasonable frozen model families or depends on selecting a favorable family after held-out access.
21. Confirmatory endpoints, margins, nulls, or thresholds were changed after the held-out seal was opened.

These are not cosmetic cautions. They define how the proposal can lose.

19. Interpretation: from Telos to conditional agency

The Telos bridge is strongest when it is treated as a disciplined question about historical constraint, not as a declaration that the universe wants an outcome. Open systems acquire structure through prior interaction. Some structures change how later perturbations are received, transformed, and acted upon. Some of those changes can be interpreted as regulation. A smaller subset may implement writable memory and consequence-sensitive adaptation. A still smaller set may satisfy the operational package for conditional agency.

The framework therefore preserves a meaningful continuum without erasing categorical obligations. Dissipation can support organized structure, as Prigogine emphasized [13], but dissipative order is not automatically control. Control can maintain a variable, as cybernetics established [1,2], but regulation is not automatically learning. Learning can make policy history-dependent, but learned control is not automatically an individual with system-relative norms. Conditional agency is a conclusion supported by a convergent causal record, not a synonym for complexity.

The framework also clarifies why metaphysical extrapolation must remain separate. A cosmic model might contain boundary conditions, memory-like state, attractors, or selection effects. None of those facts, alone or together, establishes a necessary mind, divine identity, or cosmic consciousness. Paper 75 can define the agency premise used by a later argument. It does not by itself establish a necessary mind.

20. Conclusion

Paper 75 reconstructs the strongest part of the removed SIT teleonomic material while correcting its weakest moves. The revised framework no longer subtracts incomparable quantities to manufacture an agency score. It no longer derives mismatch contraction from an unrelated potential gradient. It no longer treats dissipation, history-conditioned control, and agency as nested sets. Instead it offers typed dynamics, a finite-horizon history-conditioned objective, a state-only comparator, a multidimensional claim vector, causal interventions, thermodynamic boundaries, mandatory nulls, and explicit falsifiers. Its reference benchmark demonstrates why those distinctions matter: an analyst's history can strongly improve held-out prediction while intervention on the candidate system memory has no effect. Prediction routes a causal question; it does not answer it.

The adversarial extension adds two further requirements. Predictive history must survive plausible enrichment of the measured present, and a memory-to-policy intervention must survive an instrumented direct-path sham. In the synthetic benchmark, a hidden present variable creates a strong but collapsible history gain, while an off-target route creates a large but entirely non-memory-mediated intervention effect. The true writable-history controller survives both tests. This closes a decision-logic gap without claiming that a finite experiment can prove state completeness or perfect intervention selectivity.

The robustness extension adds a third requirement: the conclusion must survive realistic degradation of the analysis contract. In the declared synthetic construction, positive history gain survives partial target engagement, candidate and outcome missingness, rare positive outcomes, a shifted present-state distribution, and two distinct model families. The incomplete-state proxy remains equivalent to zero under missingness and imbalance and has no positive gain under the shifted distribution. The distinction matters because nonbenefit is not automatically two-sided equivalence. Frozen margins, tail rules, power, multiplicity, and one-open held-out procedures make that difference auditable.

The central scientific question is now precise: when otherwise matched present conditions are held fixed, does internally accessible history causally change control, does returned consequence rewrite that history,

and does the resulting system satisfy defensible criteria for individuality, normativity, asymmetry, and alternatives? A positive answer would support conditional agency for that system and experimental context. A negative answer would leave stabilization, dissipation, or ordinary feedback intact without inflating them into purpose.

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Author release boundary

This manuscript is a Draft 5 current manuscript for author review. The incomplete-state and direct-path confounds are closed at the declared synthetic decision-logic level by Sections 12B-12D. The imperfect-intervention, missing-data, class-imbalance, distribution-shift, and model-family stress tests are closed at that same bounded level by Sections 12E-12G. The preregistration template, construction-specific equivalence regions, power rule, multiplicity control, and held-out seal are frozen. The borrow-restricted Pittendrigh and Monod sources are used only as bibliographic historical background; no quotation or body-specific inference depends on inaccessible text. Primary-source locators, first-party genealogy, nonduplication boundaries, and the claim atlas accompany the release package.

Local manuscript, computational, citation, package, and PDF checks can establish that this candidate is more complete, more accurate, and losslessly stronger than Draft 4. They cannot substitute for Micah Blumberg’s approval or for external specialist criticism. Those are post-candidate review decisions, not reasons to mislabel completed local work as unfinished. No upload, DOI mutation, provider update, website publication, journal submission, or other public action is authorized by this manuscript or its validation files.